Problem 1: Smart Home Energy Consumption using Bayesian Networks

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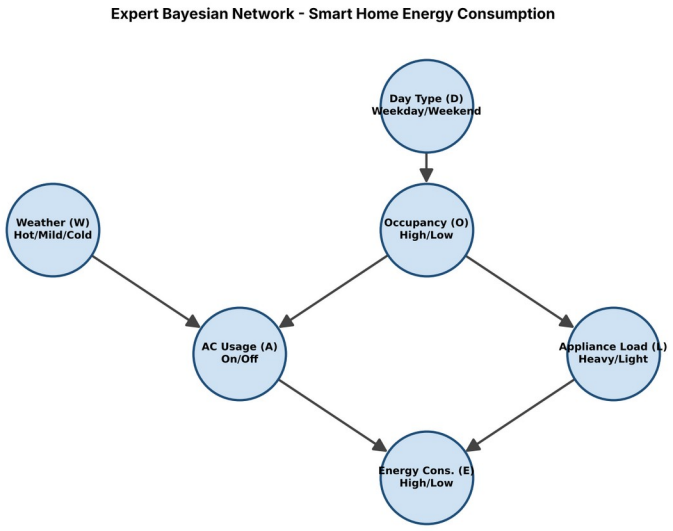
Code: Problem1_Bayesian_Network.ipynb. Screenshots: outputs folder. Random seed: 42.

Part A: Expert Bayesian Network

A.1 Network built in pgmpy with edges DayType→Occupancy, Weather→AC, Occupancy→AC, Occupancy→ApplianceLoad, AC→Energy, ApplianceLoad→Energy.

A.2 All CPTs were entered from the given tables (complements as 1 − p). The code checked all 16 parent configurations and every one sums to 1; check_model() returns True.

A.3 Network diagram:



A.4 Results:

Query	Expression	Probability
Q1	P(E=High)	0.5806
Q2	P(E=High | Weather=Hot)	0.7063
Q3	P(E=High | DayType=Weekend)	0.6402
Q4	P(E=High | Weather=Cold, Occupancy=High)	0.4843
Q5	P(Weather=Hot | E=High)	0.6082

Q6	P(AC=On | E=High)	0.6929
Q7	P(AC=On | E=High, Load=Heavy)	0.6499

A.5 Q6 = 0.6929 drops to Q7 = 0.6499. AC and appliance load are two separate causes of high consumption. If all we know is that the bill was high, the AC is a likely culprit. Once we also learn the load was heavy, that already explains most of the high bill, so we don't need the AC as much and its probability goes down. This is explaining away. The drop is small because both AC and load depend on occupancy, and a heavy load hints that people are home, which pushes AC use up a little.

A.6 Q1 by hand:

Step 1: P(O=High) = 0.7×0.45 + 0.3×0.80 = 0.555, so P(O=Low) = 0.445
Step 2: P(A=On|O=High) = 0.5×0.90 + 0.3×0.40 + 0.2×0.10 = 0.590
P(A=On|O=Low) = 0.5×0.60 + 0.3×0.15 + 0.2×0.05 = 0.355
Step 3: P(E=H|O=High) = 0.59×0.75×0.92 + 0.59×0.25×0.68
+ 0.41×0.75×0.55 + 0.41×0.25×0.12
= 0.4071 + 0.1003 + 0.1691 + 0.0123 = 0.6888
P(E=H|O=Low) = 0.355×0.35×0.92 + 0.355×0.65×0.68
+ 0.645×0.35×0.55 + 0.645×0.65×0.12
= 0.1143 + 0.1569 + 0.1242 + 0.0503 = 0.4457
Step 4: P(E=High) = 0.555×0.6888 + 0.445×0.4457 = 0.3823 + 0.1983 = 0.5806

The code gives 0.5806 too, so the two match.

Part B: Conditional Independence

Paths: (1) W→A←O←D (2) W→A→E←L←O←D (3) W→A←O→L (4) W→A→E←L

Statement	**d-separation reasoning**	**Numerical check**	**Result**
B.1 W ⊥ D	Path 1: collider A is unobserved, and so is its descendant E, so it's blocked. Path 2: collider E is unobserved, so it's blocked.	P(W=Hot) = 0.5000 P(W=Hot | D=Weekend) = 0.5000	Holds
B.2 W ⊥ D | E	E is observed, which opens collider E (path 2) and collider A through its descendant (path 1).	P(W=Hot | E=High) = 0.6082 P(W=Hot | E=High, D=Weekend) = 0.5998	Does not hold
B.3 W ⊥ L | O	Path 3: O is an observed fork, so it's blocked. Path 4: collider E is	P(W=Hot | O=High) = 0.5000 P(W=Hot | O=High, L=Heavy) = 0.5000	Holds

	unobserved, so it's blocked.		

Part C: Synthetic Data Generation

We wrote our own ancestral sampler. It visits the nodes in topological order (DayType, Weather, Occupancy, AC, ApplianceLoad, Energy) and draws each node from its CPT, given the parent values already sampled. Seeds: 42 (n=500), 43 (n=5000), 44 (n=20000).

Weather	**True**	**n=500**	**n=5000**	**n=20000**
Hot	0.50	0.4920 (246)	0.4990 (2495)	0.4960 (9921)
Mild	0.30	0.3000 (150)	0.3066 (1533)	0.3017 (6034)
Cold	0.20	0.2080 (104)	0.1944 (972)	0.2023 (4045)

Weather, Occupancy	**n=500**	**n=5000**	**n=20000**
Hot, High	145	1404	5485
Hot, Low	101	1091	4436
Mild, High	78	856	3363
Mild, Low	72	677	2671
Cold, High	65	513	2300
Cold, Low	39	459	1745

Part D: Learning from Data

D.1–D.2 For each dataset we learned two networks with the expert structure: one using MLE and one using Bayesian estimation (BDeu, ESS = 10).

D.3 P(AC=On | Weather, Occupancy), expert vs learned:

W, O	**Expert**	**MLE 500**	**Bayes 500**	**MLE 5000**	**Bayes 5000**	**MLE 20000**	**Bayes 20000**
Hot, High	0.90	0.8966	0.8920	0.9010	0.9005	0.9012	0.9011
Hot, Low	0.60	0.7129	0.7094	0.5940	0.5938	0.5877	0.5877
Mild, High	0.40	0.3462	0.3494	0.3879	0.3881	0.4059	0.4059
Mild, Low	0.15	0.1806	0.1878	0.1595	0.1604	0.1411	0.1414
Cold, High	0.10	0.0923	0.1025	0.1033	0.1046	0.1148	0.1151
Cold, Low	0.05	0.0000	0.0205	0.0545	0.0561	0.0458	0.0463

D.4 We ran Q1 to Q7 on all six learned models. The results are in the Part E tables.

Part E: Comparative Analysis

Dataset size = 500

Query	Expert Model	MLE Model	Bayesian Model	|Diff| MLE	|Diff| Bayesian
Q1	0.5806	0.5822	0.5807	0.0016	0.0001
Q2	0.7063	0.7466	0.7397	0.0404	0.0334
Q3	0.6402	0.6226	0.6181	0.0176	0.0221
Q4	0.4843	0.4370	0.4420	0.0472	0.0422
Q5	0.6082	0.6309	0.6227	0.0228	0.0145
Q6	0.6929	0.7240	0.7199	0.0312	0.0271
Q7	0.6499	0.6672	0.6647	0.0173	0.0148

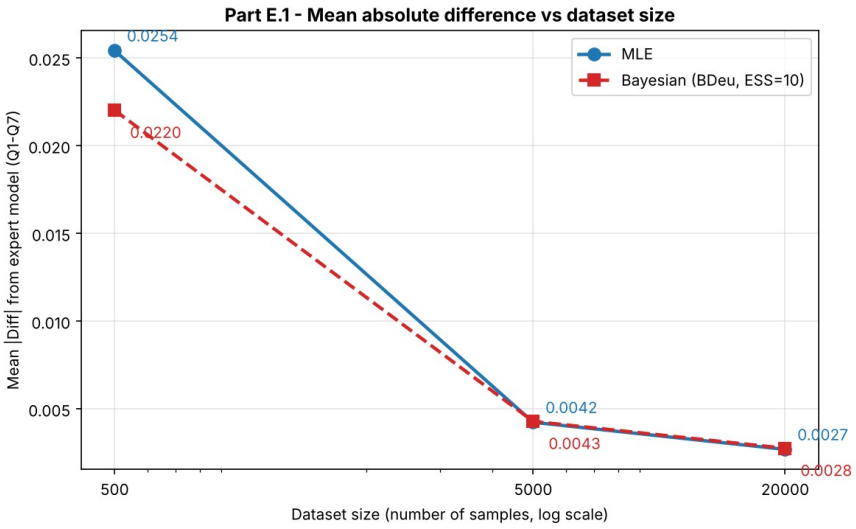
Dataset size = 5000

Query	Expert Model	MLE Model	Bayesian Model	|Diff| MLE	|Diff| Bayesian
Q1	0.5806	0.5853	0.5852	0.0047	0.0045
Q2	0.7063	0.7097	0.7091	0.0034	0.0029
Q3	0.6402	0.6453	0.6447	0.0051	0.0045
Q4	0.4843	0.4916	0.4920	0.0074	0.0078
Q5	0.6082	0.6050	0.6043	0.0031	0.0039
Q6	0.6929	0.6902	0.6899	0.0026	0.0030
Q7	0.6499	0.6466	0.6464	0.0033	0.0035

Dataset size = 20000

Query	Expert Model	MLE Model	Bayesian Model	|Diff| MLE	|Diff| Bayesian
Q1	0.5806	0.5797	0.5796	0.0009	0.0010
Q2	0.7063	0.7056	0.7054	0.0007	0.0008
Q3	0.6402	0.6425	0.6423	0.0023	0.0021
Q4	0.4843	0.4916	0.4917	0.0074	0.0075
Q5	0.6082	0.6038	0.6036	0.0044	0.0046
Q6	0.6929	0.6921	0.6920	0.0008	0.0009
Q7	0.6499	0.6475	0.6475	0.0024	0.0024

E.1 Mean |Diff| over Q1–Q7. MLE: 0.0254, 0.0042, 0.0027. Bayesian: 0.0220, 0.0043, 0.0028 (for n = 500, 5000, 20000).



E.2 Q4 has the largest error at n=500 (MLE 0.0472, Bayesian 0.0422). Its evidence, a cold day with high occupancy, showed up in only 65 of the 500 samples (13%), which is fewer than for any other query, and the AC was on in just 6 of them. With so few matching rows, the CPT entries that Q4 depends on are noisy, and that noise goes straight into the answer. Q1 has no evidence, so it averages over all the data and its errors mostly cancel out (0.0016).

E.3 For small data, Bayesian estimation does better: its mean error at n=500 is 0.0220 against 0.0254 for MLE, and it is closer on 6 of the 7 queries. MLE just copies the counts. So when a combination is rare, a few unlucky samples throw it off, and an event that never appears gets a probability of exactly zero, like P(AC=On | Cold, Low) = 0 here. The BDeu prior adds a few imaginary, evenly spread counts. These pull extreme estimates toward the middle and rule out hard zeros. With more data, those extra counts stop mattering and the two estimators give almost the same results.